

Roll No.....

END SEMESTER EXAMINATION 2025

Name of the Program: B. Tech. (Mechanical Engineering) Semester: V
Name of the Course: Solar Energy Engineering Course Code: TME511

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries equal marks.

Q1	(20 marks)	CO1
(a)	Write different laws of heat transfer. Explain overall heat transfer coefficient with suitable example.	
(b)	Define different kinds of solar radiation. What are different methods to measure solar radiation? Explain any one method with neat sketch.	
(c)	Explain latitude, declination, hour angle, zenith angle and solar altitude with neat sketch.	
Q2	(20 marks)	CO2
(a)	Explain the working of compound parabolic concentrator collector (CPC) with neat sketch. Perform thermal analysis of CPC and hence find the expression for its instantaneous efficiency.	
(b)	Explain the various performance parameters of the solar collector.	
(c)	Evaluate uniform end-of-year annual cost for a collector having present value as Rs. 15,000, salvage value as Rs. 1000, rate of interest as 5%, temperature at the outlet is 57°C and temperature at inlet is 28°C, life as 15 years, average solar intensity is 950 W/m ² and surface area of the collector is 1.0 m ² . Also, calculate thermo-economic parameter. Assume any missing data.	
Q3	(20 marks)	CO3
(a)	Explain the working of double slope passive and active solar stills with neat sketch.	
(b)	Explain the working of inverted solar still with neat sketch? Also, write its advantages and disadvantages.	
(c)	Explain the working of tubular solar still with neat sketch. What will be the effect of nanofluid and phase change material on the performance of tubular solar still.	
Q4	(20 marks)	CO4, CO5
(a)	Define a solar dryer? Write the application of solar dryer. Explain the working of indirect solar drying with neat sketch. What is its advantage over open sun drying?	
(b)	Explain different kinds of latent heat storage with neat sketch.	

(c)	Explain the basic concept of solar house. Draw and explain the characteristic curve, or I-V curve of solar panel.	
Q5	(20 marks)	CO6
(a)	A solar panel purchased for Rs. 25000 is expected to generate annual revenue of Rs. 2500 and have salvage value of Rs. 5000 at the end of 25 years. If a 8% per year required return is imposed on the purchase, compute the payback period.	
(b)	What is depreciation? Explain the different methods of calculating depreciation.	
(c)	What do you mean by capital recovery factor (CRF) and sinking fund factor (SFF)? Calculate CRF and SFF for different number of years ($n = 4$ and 8) for a given rate of interest ($i = 0.08$ and 0.10). Assume any missing data.	